

Urban Heat Islands and Vegetation Cover in Ho Chi Minh City

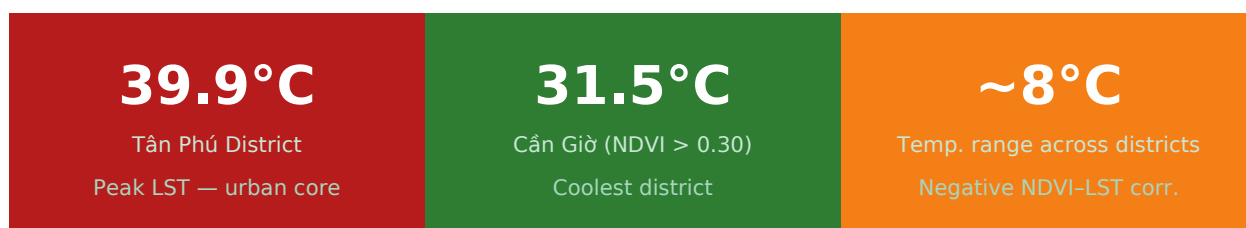
A Remote Sensing Analysis of LST and NDVI at District Level — 2024

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Tools: Google Earth Engine · Landsat 8 · Sentinel-2 · R (terra, sf, tmap, ggplot2)

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1. Introduction

The Urban Heat Island (UHI) effect is a well-documented phenomenon in which urban areas exhibit significantly higher temperatures compared to their surrounding rural regions. These temperature differences are largely the result of anthropogenic activities and the high concentration of impervious surfaces — such as buildings and roads — that absorb and retain heat. The UHI effect tends to be most pronounced under clear skies and low wind conditions, typically peaking a few hours after sunset. Its intensity can vary depending on city size and characteristics of the rural reference, with maximum temperature differences often reaching up to 10°C (Heisler & Brazel, 2010).

UHIs have been extensively studied worldwide to quantify their intensity, understand the underlying physical processes, evaluate their environmental and health impacts, and differentiate UHI-induced warming from broader global climate change driven by greenhouse gas accumulation (Heisler & Brazel, 2010). The main drivers of UHI are differences in land surface materials and urban morphology, which influence the surface energy balance and alter heat retention and radiation patterns (Heaviside et al., 2017).

Although the UHI effect can occur in any urbanized area, it is generally more intense in large metropolitan cities. In extreme cases — especially at night — city centers such as New York, London, Manchester, and Birmingham have been observed to be 5–10°C warmer than surrounding rural areas. On average, UHI intensity typically ranges from 2 to 4°C (Bohnenstengel et al., 2011; Gedzelman et al., 2003).

UHI effects can intensify environmental and health risks, particularly during heatwaves. They have been associated with altered rainfall patterns (Collier, 2006), increased air pollution levels (Xu et al., 2014), higher flood risk, and decreased water quality. The most direct public health impact arises from increased heat stress in densely populated urban zones. However, in colder seasons, some regions may benefit from slightly higher temperatures (Heaviside et al., 2017).

Ho Chi Minh City (HCMC), located in southern Vietnam, is one of the country's most populous urban centers, with over 7 million inhabitants (approximately 8.4% of Vietnam's total population as of 2012) and a high population density of approximately 2,660 inhabitants per km². According to the Köppen-Geiger classification, HCMC has a tropical wet and dry climate, with an average annual temperature of 28°C and around 1,800 mm of rainfall per year (Dang et al., 2018). The city has two distinct seasons: a rainy season from May to November and a dry season from December to April.

Given its dense population, rapid urban expansion, and significant loss of vegetative cover, Ho Chi Minh City is particularly vulnerable to the UHI effect. These factors have led to increasing surface temperatures and make the city an important case study for UHI research in Southeast Asia. Therefore, the aim of this report is to analyze the spatial patterns of surface temperature and

vegetation cover in Ho Chi Minh City during 2024, and to evaluate the relationship between NDVI and LST at the district level.

2. Objectives

- Map the spatial distribution of Land Surface Temperature (LST) across HCMC districts in 2024.
- Map the Normalized Difference Vegetation Index (NDVI) distribution as a proxy for vegetation cover.
- Quantify the relationship between NDVI and LST at the district level through regression analysis.
- Identify districts most vulnerable to the Urban Heat Island effect.
- Contribute spatial evidence to support evidence-based urban greening policies.

3. Methods

3.1 Study Area

The study focused on the administrative boundaries of Ho Chi Minh City, located in southern Vietnam. District-level boundaries (Level 2) were obtained from the Global Administrative Areas (GADM v4.1) dataset. The selected area encompasses both densely urbanized central districts and peripheral areas with varying degrees of vegetation cover.

3.2 Data Sources

Two satellite datasets were used for this analysis:

- **NDVI:** Sentinel-2 Surface Reflectance (COPERNICUS/S2_SR) via Google Earth Engine (GEE). Annual mean composite for January–December 2024. Bands B8 (NIR) and B4 (Red). Cloud filtering: <20% coverage using `MSK_CLDPRB`.
- **LST:** Landsat 8 Level 2 Surface Temperature (LANDSAT/LC08/C02/T1_L2) via GEE. Annual mean composite for January–December 2024. Band `ST_B10` (thermal), converted from Kelvin to Celsius (-273.15). Cloud pixels removed using `QA_PIXEL` bitmask.

3.3 Image Preprocessing in Google Earth Engine

In Google Earth Engine, NDVI and LST were calculated and composited into annual mean values for 2024. A user-defined rectangular polygon (roi) was drawn to cover the urban extent of HCMC. The final rasters were clipped to this region and exported at 30-meter spatial resolution to Google Drive in GeoTIFF format.

3.4 Spatial Analysis in R

The downloaded NDVI and LST rasters were processed in R 4.x using the packages *terra*, *sf*, *tmap*, *ggplot2*, and *dplyr*. The workflow consisted of five steps:

- **Reprojection:** Rasters were reprojected to EPSG:32648 (UTM Zone 48N) for accurate spatial metrics.
- **Shapefile Filtering:** The GADM Level 2 shapefile was filtered to isolate HCMC districts.
- **Zonal Statistics:** Mean NDVI and LST values were extracted per district using `terra::extract()`.

- **Visualization:** Thematic maps generated with *tmap v4*; scatterplot created with *ggplot2*.
- **Data Export:** Summary table (mean NDVI and LST per district) exported as .csv for reporting.

Component	Source / Tool	Details
NDVI	Sentinel-2 SR (GEE)	Bands B8/B4 · Annual mean 2024 · 30 m
LST	Landsat 8 L2 (GEE)	ST_B10 · Kelvin to °C · 30 m
District limits	GADM v4.1	Level 2 — Ho Chi Minh City
Preprocessing	Google Earth Engine	Cloud masking, compositing, export
Spatial analysis	R (terra, sf)	Reprojection, zonal statistics
Visualization	R (tmap v4, ggplot2)	Thematic maps, regression scatterplot

Table 1. Summary of data sources and analytical tools used in this study.

4. Results

4.1 NDVI Distribution

The spatial distribution of NDVI in Ho Chi Minh City for 2024 revealed significant heterogeneity across districts. Northern and western districts — including Củ Chi, Hóc Môn, and Bình Chánh — exhibited relatively higher NDVI values (0.25–0.35), indicating greater vegetative cover. In contrast, the central and highly urbanized districts such as District 1, District 3, and District 10 showed markedly lower NDVI values (0.08–0.11), consistent with high building density and minimal green space (Figure 1).

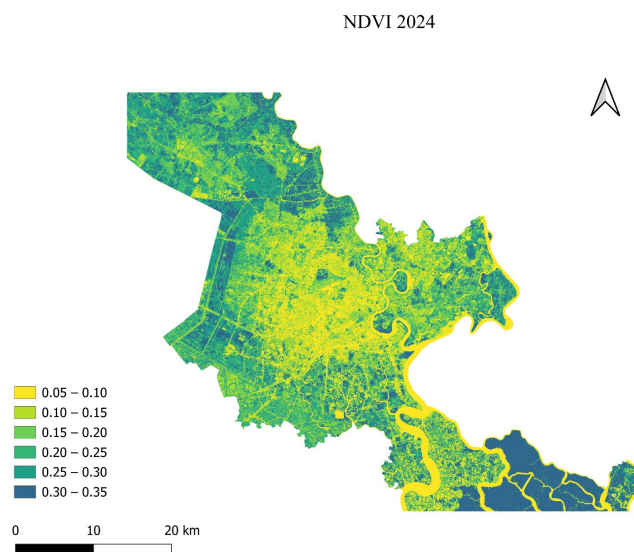


Figure 1. Spatial distribution of NDVI across Ho Chi Minh City (2024). Higher NDVI values (blue) indicate areas with denser vegetation cover, concentrated in peripheral districts. Source: Sentinel-2 / GEE. Analysis: Israel Vallejo Muñoz.

4.2 Land Surface Temperature (LST)

The mean Land Surface Temperature in 2024 ranged from 30.5°C to nearly 40°C across districts. The highest temperatures were concentrated in central and southeastern districts — corresponding to areas of high impervious surface and low vegetation cover — with Tân Phú reaching 39.97°C and District 11 reaching 39.79°C. Cooler surface temperatures were observed in districts with higher vegetation coverage, particularly in peripheral and suburban areas, with Cần Giò recording only 31.48°C (Figure 2).

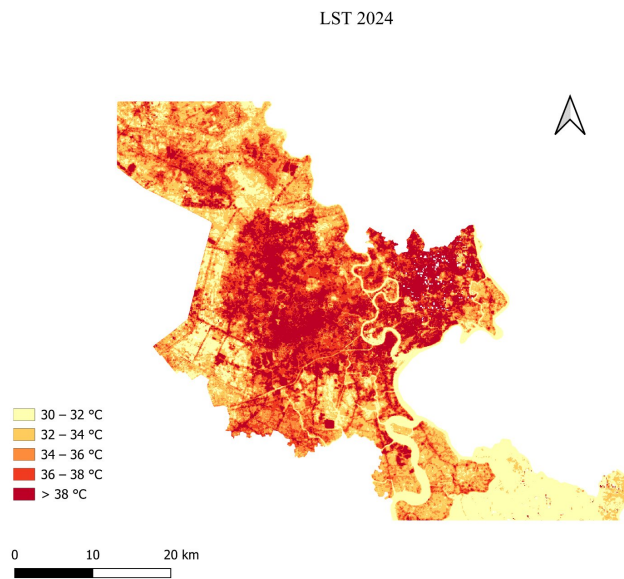


Figure 2. Spatial distribution of Land Surface Temperature (°C) across Ho Chi Minh City (2024). Warmer areas (dark red) correspond to dense urban zones with low vegetation. Source: Landsat 8 / GEE. Analysis: Israel Vallejo Muñoz.

4.3 NDVI-LST Relationship

A scatterplot analysis showed a clear negative correlation between NDVI and LST values at the district level. A linear regression model revealed that districts with higher NDVI consistently exhibited lower surface temperatures, supporting the hypothesis that vegetation plays a significant role in mitigating urban heat (Figure 3). The temperature range across districts exceeded 8°C, with the highest LST values found in the most densely built and least vegetated areas. This relationship aligns with previous UHI studies and confirms the cooling effect of vegetation in tropical urban environments like Ho Chi Minh City.

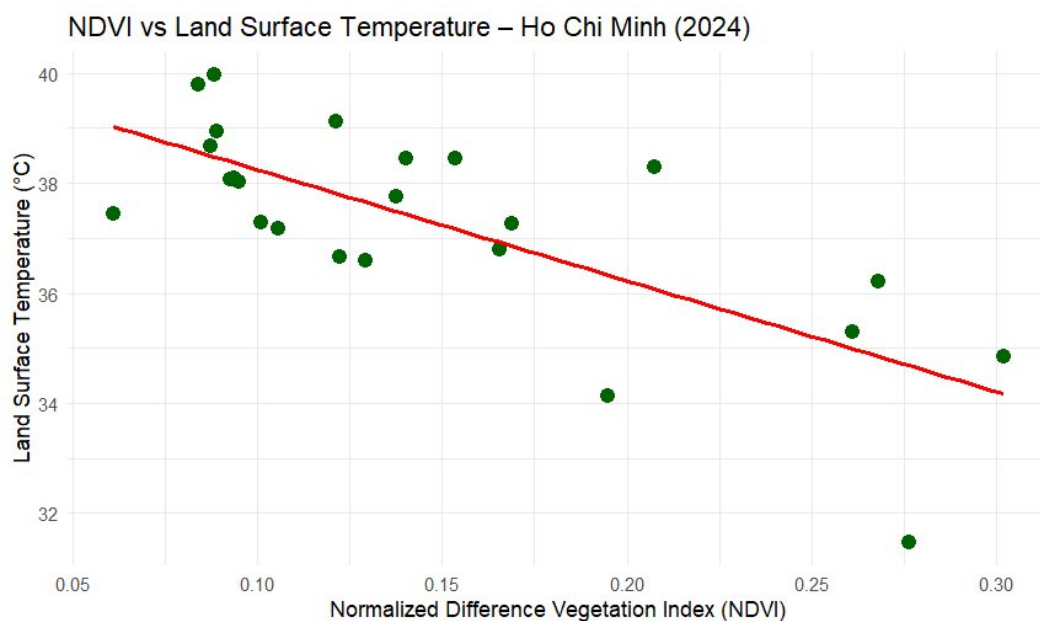


Figure 3. Scatter plot showing the inverse correlation between NDVI and Land Surface Temperature at district level — Ho Chi Minh City (2024). The red regression line confirms the negative relationship: higher vegetation density consistently predicts lower surface temperatures. Source: Sentinel-2 + Landsat 8 / GEE. Analysis: Israel Vallejo Muñoz.

4.4 District-Level Summary

Table 2 presents the mean NDVI and LST values for each administrative district of Ho Chi Minh City derived from 2024 remote sensing data, ordered from highest to lowest LST.

District	Mean NDVI 2024	Mean LST 2024 (°C)
Tân Phú	0.0882	39.98
District 11	0.0839	39.79
Tân Bình	0.1212	39.11
District 6	0.0889	38.94
District 3	0.0872	38.68
Bình Tân	0.1402	38.45
Thủ Đức	0.1536	38.44
District 9	0.2074	38.29
Phú Nhuận	0.0935	38.09
District 10	0.0949	38.02
District 5	0.0928	38.08
District 8	0.1375	37.76
District 4	0.0610	37.44
Gò Vấp	0.1008	37.30
District 12	0.1689	37.26
District 1	0.1056	37.17
District 2	0.1657	36.79
District 7	0.1225	36.67
Bình Thạnh	0.1293	36.58
Hóc Môn	0.2680	36.21
Nhà Bè	0.1949	34.13
Củ Chi	0.3020	34.84
Bình Chánh	0.2611	35.29
Cần Giờ	0.2763	31.48

Table 2. Mean NDVI and Land Surface Temperature by district in Ho Chi Minh City (2024), ordered by descending LST. Source: Sentinel-2 + Landsat 8 / GEE. Analysis: Israel Vallejo Muñoz.

5. Discussion

The findings of this study demonstrate a pronounced Urban Heat Island effect in Ho Chi Minh City, where spatial variations in land surface temperature are strongly linked to patterns of vegetation cover as measured by NDVI. The results confirm that urban districts with low vegetation cover tend to exhibit significantly higher surface temperatures, while districts with more green space maintain relatively cooler thermal profiles.

This pattern is in line with well-established ecological principles and has been observed in numerous cities worldwide, including both temperate and tropical climates (Heaviside et al., 2017; Bohnenstengel et al., 2011). Vegetation serves as a critical regulator of surface temperatures by providing shade, increasing evapotranspiration, and altering the surface energy balance. Urban areas lacking vegetation tend to absorb and retain heat more efficiently due to the presence of impervious materials such as asphalt, concrete, and metal, which have high thermal inertia and low albedo.

Key Finding: Central districts such as Tân Phú (39.97°C) and District 11 (39.79°C) — with NDVI values below 0.09 — recorded the highest surface temperatures. In contrast, outer districts such as Cần Gière (NDVI = 0.276, LST = 31.48°C) and Củ Chi (NDVI = 0.302, LST = 34.84°C) remained significantly cooler, with temperature differences exceeding 8°C across the city.

The UHI effect in HCMC is particularly concerning due to the city's tropical climate, which already predisposes the population to heat stress. When combined with climate change and increased frequency of heatwaves, the UHI effect can exacerbate environmental and public health risks. Studies have shown that elevated nighttime temperatures — often intensified by UHI conditions — can impair sleep quality, increase cardiovascular stress, and raise mortality rates, especially among vulnerable populations such as the elderly, low-income communities, and outdoor workers (Heaviside et al., 2017; Xu et al., 2014).

Beyond health, urban heat also has implications for energy consumption and infrastructure resilience. Higher ambient temperatures drive increased demand for air conditioning, placing strain on electrical grids and increasing greenhouse gas emissions in a feedback loop. Road surfaces and buildings also degrade more rapidly under sustained heat exposure, leading to higher maintenance costs.

Interestingly, while the inverse correlation between NDVI and LST was evident and statistically consistent, the relationship is not perfectly linear. Some districts with moderate NDVI levels still experienced elevated temperatures, which may be attributed to urban geometry, building height (which affects radiative trapping), and local anthropogenic emissions such as traffic and industry. This suggests that UHI mitigation strategies should not rely solely on increasing green cover, but must be part of a broader framework including urban design, ventilation corridors, sustainable transportation, and building material selection.

The results of this study closely align with the key findings of Dang et al. (2018), who quantified the influence of green space on UHI-attributable mortality in HCMC. Both approaches converge in showing that districts with limited vegetation are systematically exposed to higher thermal stress. Dang et al. estimated that up to 24% of heat-related deaths in HCMC could be prevented with adequate green infrastructure coverage — our spatial data provide the environmental mechanism

supporting their conclusion.

Importantly, while Dang et al. relied on MODIS-based NDVI at 250 m resolution, this study utilized Landsat 8 at 30 m, providing finer intra-urban variation. Despite this methodological difference, the overarching message is consistent: vegetation is fundamental to reducing thermal exposure, and its uneven distribution creates disparities in climate-related health risks across the city.

Furthermore, Nguyen et al. (2023) showed that NDVI values in HCMC's urban districts have been steadily decreasing from 2013 to 2022 — particularly in central and southern areas — while LST has increased correspondingly. Our 2024 snapshot confirms that central districts remain ecological hotspots ($\text{NDVI} < 0.15$, $\text{LST} > 36^{\circ}\text{C}$), consistent with the long-term trend identified by Nguyen et al. Together, these results strongly advocate for evidence-based urban policies that prioritize green infrastructure.

One notable limitation of this study is the reliance on annual composites of satellite imagery. While this approach minimizes cloud contamination, it may underrepresent short-term temperature peaks caused by seasonal variability. Future analyses should incorporate seasonal or monthly composites to better capture heat event dynamics. Additionally, integrating high-resolution sensors such as ECOSTRESS and socio-demographic data could enable neighborhood-scale assessments and equity-centered adaptation planning.

6. Conclusions

This study demonstrates the strong inverse relationship between urban vegetation (NDVI) and land surface temperature (LST) across the administrative districts of Ho Chi Minh City in 2024, confirming the presence and spatial heterogeneity of the Urban Heat Island effect. Areas with sparse vegetation and dense urban infrastructure — such as Tân Phú, District 11, and District 3 — exhibited surface temperatures consistently higher than those in more vegetated outer districts (Củ Chi, Cần Giẽ), with differences exceeding 8°C in some cases.

By leveraging high-resolution satellite imagery and district-level spatial analysis, this work complements and reinforces the conclusions of Dang et al. (2018), who found that reduced green space in HCMC contributes to excess mortality due to heat stress. While their research focused on epidemiological implications, our geospatial analysis elucidates the environmental mechanisms underlying those risks.

The findings highlight the unequal distribution of green cover across the city and its environmental consequences, pointing to the need for targeted urban planning interventions. Expanding vegetated areas — especially in vulnerable inner-city districts — could help mitigate the UHI effect, reduce energy demands for cooling, and enhance overall urban resilience to climate change.

Future research should integrate socio-demographic data, land-use classifications, and temporal trends to build a more comprehensive urban heat vulnerability assessment. Meanwhile, city authorities and stakeholders should consider both spatial and health-based evidence when designing adaptive strategies to combat urban overheating.

7. References

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